Point-to-Point OSPF Networks

EXT

The network Command Syntax

```
Router(config-router)# network network-address wildcard-mask area area-id
```

- The network address wildcard mask operate in exact a mode OSPF on interfaces. Any interface in a router that match the network address in the **network** command are enabled to send and receive OSPF packets.
- The area area-id specifies the area ID for the area being assigned to running single area OSPFv2. The **network** command must be configured with the same area id value on all routers. Although any area ID can be used, it is good practice to use an area ID of 0 with single area OSPFv2. This convention makes it easier if the network is later altered to support multiarea OSPFv2.

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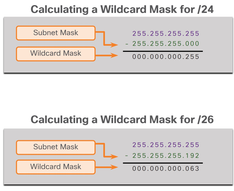
The Wildcard Mask

The wildcard mask is typically the inverse of the subnet mask configured on that interface. In a subnet mask, binary 1 is equal to a match and binary 0 is not a match. In a wildcard mask, the reverse is true, as shown in here:

- **Wildcard mask bit 0** – Matches the corresponding bit value in the address.
- **Wildcard mask bit 1** – Ignores the corresponding bit value in the address.

The easiest method for calculating a wildcard mask is to subtract the network subnet mask from 255.255.255.255, as shown for /24 and /26 subnet masks in the figure.



EXT

Check Your Understanding – The Wildcard Masks

Calculate the subnet mask and wildcard mask required to advertise the specified network address in OSPF. Type your answers in the fields provided. Click Check to verify your answers. Click Show Me to see the correct answer.

Click New Problem to continue the activity.

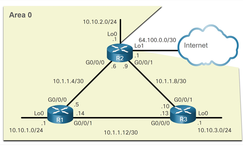
Network / Subnet / Host	10.0.0.0/8
Decimal Subnet Mask	255.0.0.0
Wildcard Mask	0.255.255.255

Check | New Problem | Show Me

EXT

Configure OSPF Using the network Command

Within routing configuration mode, there are two ways to identify the interfaces that will participate in the OSPFv2 routing process. The figure shows the reference topology.



In the first example, the wildcard mask identifies the interface based on the network addresses. Any active interface that is configured with an IPv4 address belonging to that network will participate in the OSPFv2 routing process.

```
R1(config)# router ospf 10
R1(config-router)# network 10.10.1.0 0.0.0.255 area 0
R1(config-router)# network 10.1.1.4 0.0.0.3 area 0
R1(config-router)# network 10.1.1.12 0.0.0.3 area 0
R1(config-router)#
```

Note: Some IOS versions allow the subnet mask to be entered instead of the wildcard mask. The IOS then converts the subnet mask to the wildcard mask format.

As an alternative, the second example shows how OSPFv2 can be enabled by specifying the exact interface IPv4 address using a quad zero wildcard mask. Entering **network 10.1.1.6 0.0.0.0 area 0** in R1 tells the router to enable interface Gigabit Ethernet 0/0/0 for the routing process. As a result, the OSPFv2 process will advertise the network that is on this interface (10.1.1.4/30).

```
R1(config)# router ospf 10
R1(config-router)# network 10.10.1.0 0.0.0.255 area 0
R1(config-router)# network 10.1.1.6 0.0.0.0 area 0
R1(config-router)# network 10.1.1.12 0.0.0.3 area 0
R1(config-router)#
```

The advantage of specifying the interface is that the wildcard mask calculation is not necessary. Notice that in all cases, the **area** argument specifies area 0.

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Syntax Checker – Configure R2 and R3 Using the network Command

Use the Syntax Checker to alternate the networks connected to R2 and R3.

Note: While completing the syntax checker, observe the informational messages describing the adjacency between R1 (1.1.1.1) and R2 (2.2.2.2). The IPv4 addressing scheme used for the router ID makes it easy to identify the neighbor.

You are currently logged into R2:

- Enter OSPF router configuration mode using process ID 10
- Configure the R2 router ID of 2.2.2.2

```
R2(config)# router ospf 10
R2(config-router)#
```

Reset | Show Me | Show All

EXT

Configure OSPF Using the ip ospf Command

You can also configure OSPF directly on the interface instead of using the **network** command. To configure OSPF directly on the interface, use the **ip ospf** interface configuration mode command. The syntax is as follows:

```
Router(config-if)# ip ospf process-id area area-id
```

For R1, remove the network commands by using the **no** form of the **network** commands. And then go to each interface and configure the **ip ospf** command, as shown in the command sections.

```
R1(config)# router ospf 10
R1(config-router)# no network 10.10.1.0 0.0.0.255 area 0
R1(config-router)# no network 10.1.1.4 0.0.0.3 area 0
R1(config-router)# no network 10.1.1.12 0.0.0.3 area 0
R1(config-router)# interface GigabitEthernet 0/0/0
R1(config-if)# ip ospf 10 area 0
R1(config-if)# interface GigabitEthernet 0/0/1
R1(config-if)# ip ospf 10 area 0
R1(config-if)# interface Loopback 0
R1(config-if)# ip ospf 10 area 0
R1(config-if)#
```

EXT

Syntax Checker – Configure R2 and R3 Using the ip ospf Command

Use the Syntax Checker to alternate the networks by configuring the interfaces for OSPF on R2 and R3.

You are currently logged into R3. The network commands are already removed. Configure OSPF routing using process ID 10. In area 0, on each interface, in that order, (see the following enclosed interface names):

- lo
- g0/0/0
- g0/0/1

```
R3(config)# interface lo0
R3(config-if)#
```

Reset | Show Me | Show All

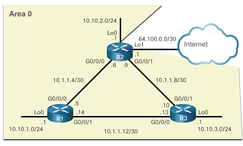
EXT

Passive Interface

By default, OSPF messages are forwarded out all OSPF-enabled interfaces. However, these messages really only need to be sent out interfaces that are connecting to other OSPF-enabled routers.

Refer to the topology in the figure. OSPFv2 messages are forwarded out the three loopback interfaces even though no OSPFv2 neighbor exists on these simulated LAN interfaces. Routing updates broadcasts received on physical interfaces to network with users and traffic. Sending out unneeded messages on a LAN affects the network in three ways, as follows:

- **Inefficient Use of Bandwidth** – Available bandwidth is consumed transporting unnecessary messages.
- **Inefficient Use of Resources** – All devices on the LAN must process and eventually discard the message.
- **Increased Security Risk** – Without additional OSPF security configurations, OSPF messages can be intercepted with packet sniffing software. Routing updates can be modified and sent back to the router, corrupting the routing table with false metrics that misdirect traffic.



EXT

Configure Passive Interfaces

Use the **passive-interface** router configuration mode command to prevent the transmission of routing messages through a router interface, but still allow that network to be advertised to other routers. The configuration example identifies the R1 Loopback 0/0/0 interface as passive.

Note: The loopback interface is this example is representing an Ethernet network. In production networks, loopback interfaces do not require to be passive.

The **show ip protocols** command is then used to verify that the Loopback 0 interface is listed as passive. The interface is still listed under the heading, "Routing on Interfaces Configured Explicitly (Area 0)", which means that the network is still included as a route entry in OSPFv2 updates that are sent to R2 and R3.

```
R1(config)# router ospf 10
R1(config-router)# passive-interface loopback 0
R1(config-router)# end
R1#
*May 15 20:34:20.389: %SYS-5-CONFIG_I: Configured from console by console
R1# show ip protocols
*** IP Routing is NSF aware ***
Routing protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
  Routing on Interfaces Configured Explicitly (Area 0):
    Loopback0
    GigabitEthernet0/0/1
    GigabitEthernet0/0/0
  Passive Interface(s):
    Loopback0
  Routing Information Sources:
    Gateway         Distance      Last Update
    2.2.2.2         110      01:01:40
    3.3.3.3         110      01:01:30
  Distance: (default is 110)
R1#
```

EXT

Syntax Checker – Configure R2 and R3 Passive Interfaces

Use the Syntax Checker to configure the Loopback interfaces on R2 as a passive. As an alternative, all interfaces can be made passive using the **passive interface default** command. Interfaces that should not be passive can be re-enabled using the **no passive-interface** command. Configure R2 with the **passive-interface default** command and then to enable the Gigabit Ethernet interface.

You are currently logged into R2:

- Enter OSPF router configuration mode using process ID 10.
- Configure the Loopback interface as passive using the shortest interface name **lo0**
- Notice no g0/0/0 and g0/0/1 exists
- Verify the OSPF settings with the **show ip protocols** command.

```
R2(config)# router ospf 10
R2(config-router)#
```

Reset | Show Me | Show All

EXT

OSPF Point-to-Point Networks

By default, Cisco routers elect a DR and BDR on Ethernet interfaces, even if there is only one other device on the link. You can verify this with the **show ip ospf interface** command, as shown in the example in R1(OSPF 10).

```
R1# show ip ospf interface GigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Internet Address 10.1.1.5/30, Interface ID 14, Area 0
  Process ID 10, Router ID 1.1.1.1, Network type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State BDR, Priority 1
                    Hello  Dead   Wait   Retransmit   Subnet
  Timer intervals   10     40     40     5            mask
  Hello due in 00:00:08
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2 (Designated Router)
  Suppress hello for 0 neighbor(s)
R1#
```

R1 is the BDR and R2 is the DR. The DR/ BDR election process is unnecessary as there can only be two routers on the point-to-point network between R1 and R2. Configure the OSPF interface on R1 and R2 to operate as a point-to-point network using the **ip ospf network point-to-point** interface command. Use the **show ip ospf network point-to-point** command on all interfaces where you want to disable the DR/BDR election process. The example below shows the configuration for R1. The OSPF neighbor adjacency status will go down for a few milliseconds.

```
R1(config)# interface GigabitEthernet 0/0/0
R1(config)#ip ip ospf network point-to-point
*May 15 00:00:20.391: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on GigabitEthernet0/0/0 from full to DOWN, Neighbor Down: Interface down or
detached
*May 15 00:00:20.391: %OSPF-5-ADJCHG: Process 10, Nbr 3.3.3.3 on GigabitEthernet0/0/0 from LOADING to FULL, Loading Done
R1(config)#int g0/0/0
R1(config-if)# ip ospf network point-to-point
*May 15 00:00:50.391: %OSPF-5-ADJCHG: Process 10, Nbr 3.3.3.3 on GigabitEthernet0/0/0 from full to DOWN, Neighbor Down: Interface down or
detached
*May 15 00:00:50.391: %OSPF-5-ADJCHG: Process 10, Nbr 3.3.3.3 on GigabitEthernet0/0/0 from LOADING to FULL, Loading Done
R1(config-if)# show ip ospf interface GigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Internet Address 10.1.1.5/30, Area 0, Attached via Interface Enable
  Process ID 10, Router ID 1.1.1.1, Network type POINT_TO_POINT, Cost: 1
                    Hello  Dead   Wait   Retransmit   Subnet
  Timer intervals   10     40     40     5            mask
                                                      Hello
  Enabled by interface config, including secondaries, ip address
  Transmit Delay is 1 sec, State POINT_TO_POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:05
    Neighbor Count is 1
    Adjacent Neighbor count is 1
    Adjacent with neighbor 2.2.2.2
    Suppress hello for 0 neighbor(s)
R1(config-if)#
```

Notice that the Gigabit Ethernet 0/0/0 interface now lists the network type as POINT_TO_POINT and that there is no DR or BDR on the link.

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Loopbacks and Point-to-Point Networks

We use loopbacks to provide additional interface for a variety of purposes. In this case, we are using loopbacks to simulate more networks than the equipment can support. By default, loopback interfaces are advertised as /32 host routes. For example, R1 would advertise the /32 / 32 host networks as 10.10.1.1/32 to R2 and R3.

```
R2# show ip route | include 10.10.1.1
O     10.10.1.1/32 [110/1] via 10.1.1.1, 00:00:01, GigabitEthernet0/0/0
```

R2 connects to and R3 has a Loopback 0 interface is configured as a point-to-point network as that R1 will advertise the full /16 / 16 network to R2 and R3.

```
R1(config)#int lo0 ip ospf network point-to-point
R1(config)#int g0/0/0 ip ospf network point-to-point
```

Now R2 receives the more accurate, advertised LAN network address of 10.10.1.0/24.

```
R2# show ip route | include 10.10.1
O     10.10.1.0/24 [110/1] via 10.1.1.1, 00:00:01, GigabitEthernet0/0/0
```

EXT

Packet Tracer – Point-to-Point Single-Area OSPFv2 Configuration

In this Packet Tracer activity, you will configure the single area (SAOPv2) with the following:

- Explicitly configure router IDs.
- Configure the **network** command on R1 using wildcard mask based on the subnet mask.
- Configure the **network** command on R2 using a quad zero wildcard mask.
- Configure the **ip ospf** command on R3.
- Configure passive interfaces.
- Verify OSPF operation using the **show ip protocols** and **show ip route** commands.

Point-to-Point Single-Area OSPFv2 Configuration

Point-to-Point Single-Area OSPFv2 Configuration